

Figure 4.1 Dapp stack

> **DEFINITION** *Dapps* are web or enterprise applications that contain decentralized smart contract logic to invoke blockchain functions.

Figure 4.2 depicts two nodes connected by a blockchain network. If you were to separate the second layer of the Dapp stack (figure 4.1) in these two nodes from its surrounding layers (as indicated by the dotted lines in figure 4.2) and take a peek, you'd see the APIs, ports, server code, and other scripts integrating the layers. These components are the ones you will work on while developing a Dapp.

Now let's follow the flow in the architectural model shown in figure 4.2. Starting at the top, the users (actors) or processes acting on behalf of users invoke the UI functions. These functions use web application software and blockchain APIs to connect to smart contract functions. Txs representing the smart contract function invocations are recorded on the blockchain. (Note that only some of the necessary Txs will be recorded.) You can follow the operational flow in a node from an actor to the consistent blockchain recording on both nodes via the blockchain network. This figure also illustrates how a blockchain-based Dapp is not a standalone application but is dependent on its host operating system's file system, ports, and network capabilities.